

THE FUTURE IS INTELLIGENT

INTRODUCTION TO AI

Explore the world of Artificial Intelligence and how it is transforming the way we live, work and innovate.



INTELLIGENT
SYSTEMS



MACHINE
LEARNING



DEEP
LEARNING



REAL WORLD
IMPACT

AI
ARTIFICIAL
INTELLIGENCE



What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the ability of a computer or machine to perform tasks that normally require **human intelligence**.

These tasks include:



Learning



Problem-solving



Understanding language



Making decisions



Recognizing images



Predicting outcomes



Simple Definition

Artificial Intelligence is the technology that enables machines to **think, learn, and make decisions** similar to humans.



Human Intelligence vs Artificial Intelligence

 Human Intelligence	 Artificial Intelligence
 Learns from experience	 Learns from data
 Uses emotions	 Uses algorithms
 Can be creative naturally	 Generates content based on patterns
 Gets tired	 Works continuously
 Makes subjective decisions	 Makes data-driven decisions

Rule-Based Systems vs Learning Systems

Before modern AI, most software followed fixed rules.



Rule-Based Systems

These systems only do exactly what they are programmed to do.

Example

```
IF age >= 18
    Allow voting
ELSE
    Reject
```

Characteristics

- Fixed rules
- No learning
- Predictable
- Easy to build
- Cannot improve automatically

Examples



Calculator



ATM menu



Traffic signal controller



Tax calculation software



Learning Systems (Modern AI)

Instead of writing every rule manually, we provide data. The AI learns patterns from that data.

Example

Instead of writing rules for spam emails:

- Contains "Lottery" → Spam
- Contains "\$\$\$" → Spam

We train AI using millions of spam and non-spam emails.

Now it learns:

- ✓ writing style
 - ✓ sender behavior
 - ✓ keywords
 - ✓ links
 - ✓ attachments
- without us programming every rule.

Characteristics

- Learns from data
- Improves over time
- Can make predictions
- Handles complex problems

Examples



Netflix recommendations



Face Unlock



Google Translate



ChatGPT



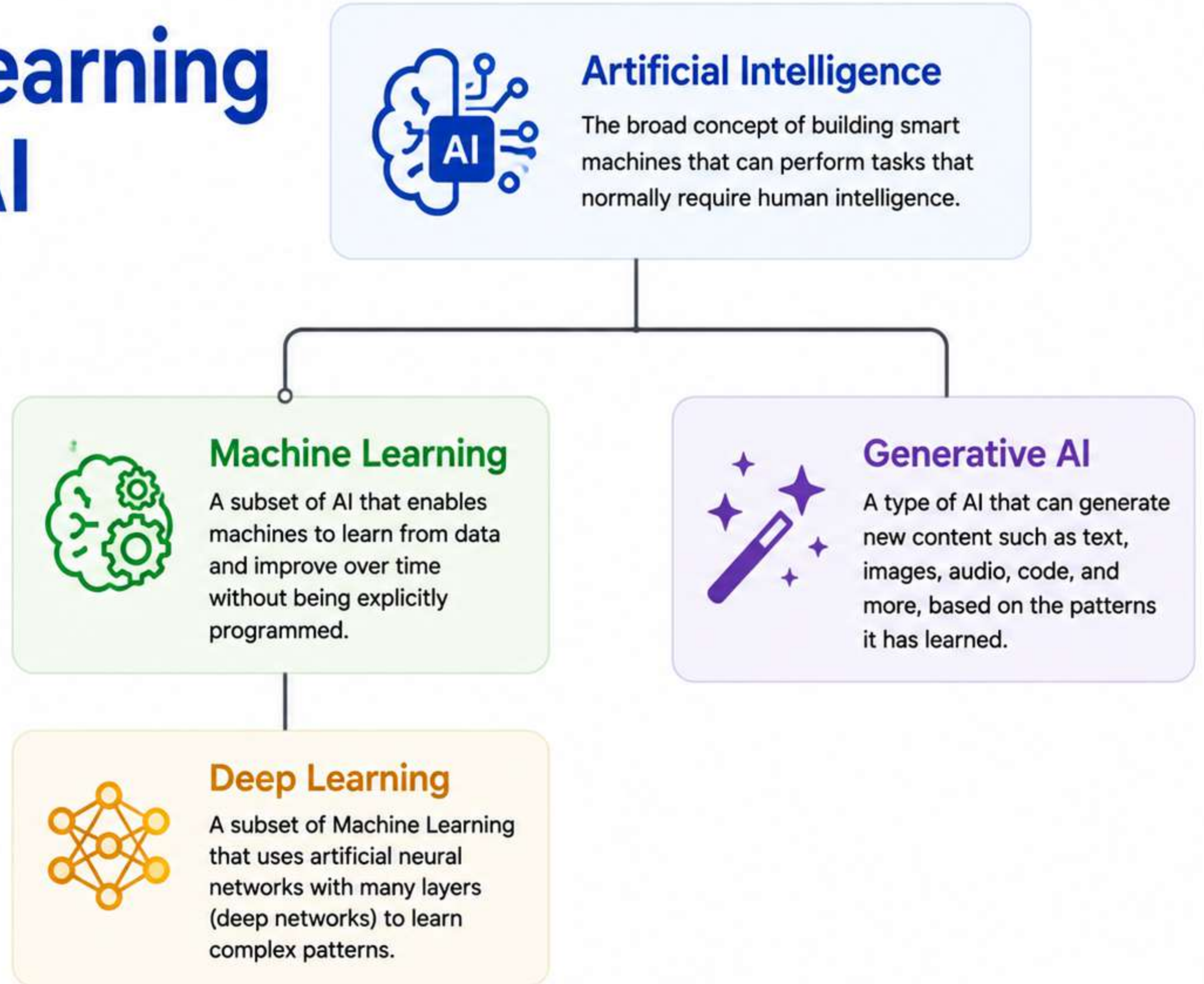
Fraud Detection

Rule-Based vs Learning Systems

 Rule-Based	 Learning System
 Fixed rules	 Learns patterns
 No improvement	 Improves with data
 Manual programming	 Model training
 Best for simple tasks	 Best for complex tasks
 Predictable	 Adaptive

AI vs Machine Learning vs Generative AI

Many people use these terms interchangeably, but **they are different.**



Artificial Intelligence (AI)

The broad field of creating intelligent machines.



Goal:

Make computers perform intelligent tasks.



Examples



Chess programs



Voice assistants



Navigation apps



Recommendation
systems

Machine Learning (ML)



A subset of AI.



Instead of programming rules, machines **learn from data**.



Example



Bank fraud detection

Input

Millions of transaction records



Machine Learning Model

Output



Fraud



Not Fraud

Applications



Credit scoring



Sales prediction



Customer churn prediction



Demand forecasting

Generative AI (GenAI)

A special type of AI that creates new content.

It generates:



Text



Images



Audio



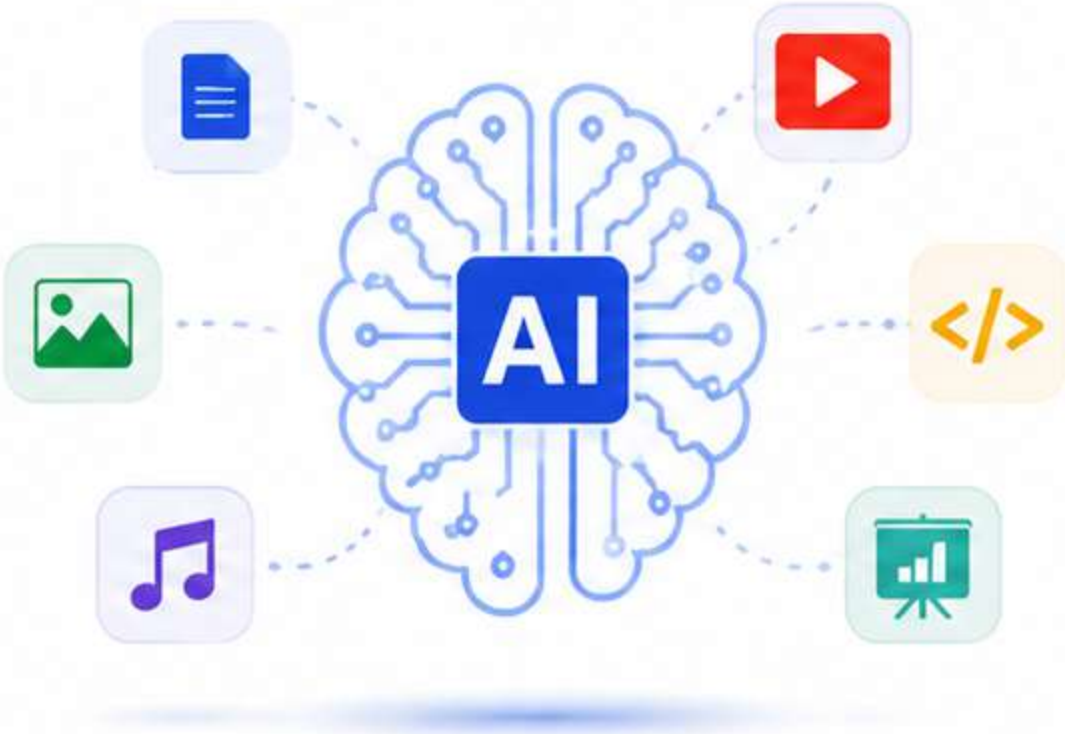
Videos



Code



Presentations



Examples

Prompt

Write an apology email



Output

A complete professional email.

Subject: Sincere Apology

Dear [Customer Name],

I sincerely apologize for the inconvenience caused. We understand your concern and take full responsibility for the issue.

We are taking immediate steps to resolve this and ensure it does not happen again.

Thank you for your understanding.

Best regards,
[Your Name]



Prompt

Create a logo



Output

An AI-generated image.



Popular GenAI tools

 ChatGPT

 Gemini

 Claude

 Microsoft Copilot

AI vs ML vs GenAI

 AI	 Machine Learning	 Generative AI
 Broad concept	 Learns from data	 Creates new content
 Includes ML	 Subset of AI	 Often built on advanced ML
 Decision making	 Prediction	 Content generation

TYPES OF AI



There are three commonly discussed types of AI based on capability.

1 NARROW AI (WEAK AI)



Designed to perform a specific task.
Cannot do anything outside
its limited scope.

EXAMPLES

- Siri, Alexa (Voice Assistants)
- Self-driving Car Systems
- Spam Filters
- Recommendation Systems (Netflix, YouTube)



Capability Level: Low

2 GENERAL AI (AGI)



AI with human-level intelligence.
Can understand, learn and apply
knowledge across different tasks
just like a human.

EXAMPLES (NOT YET ACHIEVED)

- Solve any problem a human can
- Learn new skills automatically
- Adapt to new environments
- Reason, plan and make decisions like humans



Capability Level: Medium

3 SUPER AI



AI that surpasses human intelligence
in every aspect – creativity, decision
making, problem solving, and more.

EXAMPLES (HYPOTHETICAL)

- Smarter than the best human in every field
- Solves complex global problems
- Scientific discoveries beyond human imagination
- Self-improving and self-evolving



Capability Level: Very High



Key Takeaway: Today, all AI systems we use are **Narrow AI**.

COMPARISON: TYPES OF AI



Narrow AI, General AI, and Super AI differ in capabilities and existence.

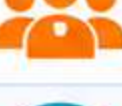
 Feature	 NARROW AI (WEAK AI)	 GENERAL AI (AGI)	 SUPER AI
 Exists Today	 Yes	 No	 No
 Single Task	 Yes	 No	 No
 Human-Level Intelligence	 No	 Yes (At human level)	 Beyond Humans
 Learns New Skills Easily	 Limited (Specific training required)	 Yes (Can learn like humans)	 Yes (Learns and improves rapidly)



Key Takeaway: Today, all AI systems we use are **Narrow AI**.

AI ACROSS INDUSTRIES

AI is transforming nearly every sector.

INDUSTRY		AI USE CASES
	Healthcare	• Disease diagnosis, medical imaging, drug discovery
	Banking	• Fraud detection, credit scoring, chatbots
	Retail	• Product recommendations, inventory forecasting
	Manufacturing	• Predictive maintenance, quality inspection
	Education	• Personalized learning, AI tutors
	Marketing	• Content creation, customer segmentation
	Human Resources	• Resume screening, interview scheduling
	Customer Service	• AI chatbots, automated support
	Logistics	• Route optimization, demand forecasting
	Agriculture	• Crop monitoring, yield prediction

Why AI Matters for Business Today

Businesses use AI because it helps them:



1

Save Time

Automates repetitive work.

Example



Writing emails, meeting summaries, report generation.



2

Reduce Costs

Less manual effort.

Example



Customer support chatbots.



3

Better Decision Making

AI analyzes huge amounts of data.

Example



Sales forecasting.



4

Improve Customer Experience

Personalized recommendations.

Example



Amazon product suggestions.



5

Increase Productivity

Employees focus on strategic work while AI handles routine tasks.



Key Takeaway: AI is not just a technology trend—it's a **business advantage**.

REAL BUSINESS EXAMPLES

AI is already powering some of the world's leading companies.

 COMPANY	 AI APPLICATION
	 Product recommendations
	 Movie recommendations
	 Search ranking and translation
	 Route optimization and pricing
	 Copilot for productivity
 Banks	 Fraud detection
 Hospitals	 Medical image analysis



Key Takeaway: Businesses of all sizes and industries are using AI to **improve efficiency**, **reduce costs**, and **deliver better experiences**.



ACTIVITY

Compare ChatGPT and Gemini



Business Problem

Draft a professional apology email to a customer whose order was delayed by **three days**.



Instructions

- 1 Work in pairs.
- 2 Use the same prompt in both **ChatGPT** and **Gemini**.
- 3 Compare the outputs based on:



Tone



Clarity



Professionalism



Personalization



Length



Comparison Criteria

Criteria	 ChatGPT	 Gemini
Tone		
Clarity		
Professionalism		
Personalization		
Length		



Discuss:

- Which response is more useful?
- What differences did you notice?
- How could the prompt be improved?



Learning Outcome:

Participants see that different AI tools can produce different results from the same prompt, highlighting the importance of prompt quality and critical evaluation.

